Customer Shopping Behavior Analysis

Retail Analytics Portfolio Report

Python · MySQL · Power BI

Executive Summary

This project analyzes 3,900 retail purchase records to understand the customer, product, and behavioral factors associated with sales and repeat purchasing. The analysis combines Python-based data preparation and exploratory analysis, MySQL business queries, and an interactive Power BI dashboard.

The customer records represent **$233,081 in observed revenue**, with an **average purchase amount of $59.76** and an **average review rating of 3.75**. Clothing was the largest revenue category, while Young Adults generated the highest age-group revenue. Discounts, subscriptions, ratings, and previous-purchase counts showed limited association with current transaction value, suggesting that the retailer should prioritize targeted testing over broad assumptions.

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1. Business Problem

A leading retail company convened its C-suite to understand changes in customer shopping behavior and determine how those changes could be used to improve sales, customer satisfaction, and long-term loyalty.

Management wanted to understand how purchasing patterns varied across demographics, product categories, and shopping preferences. The team was particularly interested in whether discounts, review ratings, seasons, subscriptions, payment methods, shipping preferences, and purchase history influenced customer decisions and repeat purchases.

> **Business question:** How can the company leverage customer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Analysis Objectives

- Identify the customer and product segments contributing the most revenue.
- Examine the relationship between discounts, ratings, subscriptions, and purchase behavior.
- Evaluate repeat-purchase patterns and customer lifecycle opportunities.
- Translate the findings into practical marketing, merchandising, and loyalty actions.

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2. Dataset Overview

Dataset characteristic	Value
Purchase records	3,900
Original columns	18
Product categories	4
Unique products	25
Customer locations	50
Missing review ratings	37

The dataset contains customer demographics, product details, purchase amounts, seasons, review ratings, subscription status, shipping choices, discounts, previous purchases, payment methods, and stated purchase frequency.

Data Preparation

- Loaded and audited the source data in Python using pandas.
- Checked data types, distributions, missing values, and duplicate rows.
- Imputed 37 missing review ratings using the median rating within each category.
- Standardized column names to snake case.
- Created age-group and purchase-frequency-day features.
- Removed `promo_code_used` because it duplicated `discount_applied`.
- Exported the cleaned dataset for MySQL and Power BI.

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3. Analytical Approach

Python: Exploration and Cleaning

Python was used to inspect data quality, prepare the analysis-ready dataset, calculate descriptive metrics, and explore relationships among customer and purchase attributes.

MySQL: Business Analysis

SQL queries addressed revenue by customer group, high-spending discount users, top-rated products, shipping behavior, subscription performance, discount dependence, lifecycle segments, category leaders, repeat buyers, and age-group revenue.

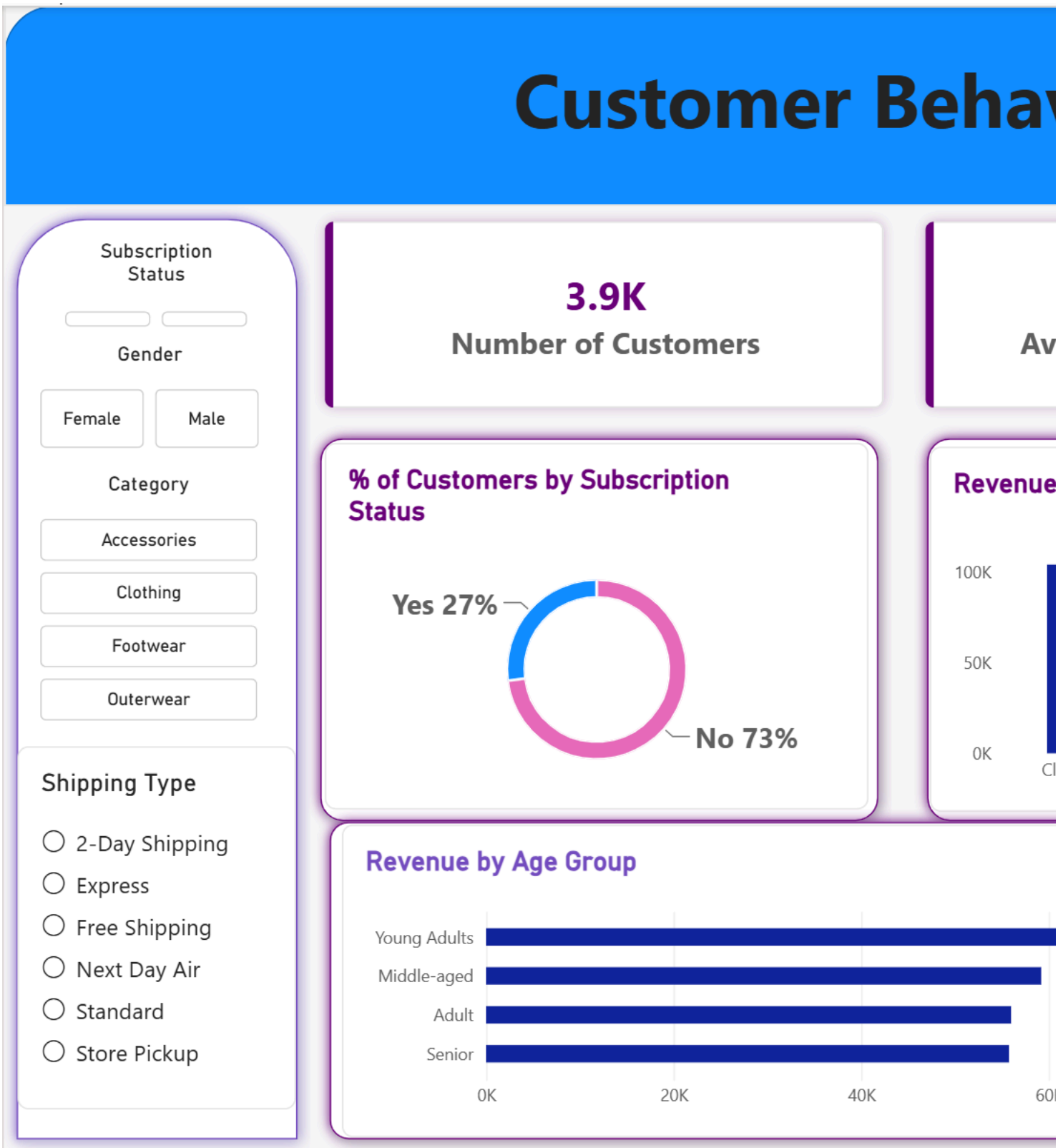
Power BI: Interactive Reporting

Power BI was used to create KPI cards, category and age-group comparisons, subscription analysis, and interactive filters for gender, category, shipping type, and subscription status.

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4. Performance Snapshot

KPI	Result
Total observed revenue	**$233,081**
Average purchase amount	**$59.76**
Average review rating	**3.75 / 5**
Subscriber share	**27%**
Clothing revenue	**$104,264**
Clothing share of revenue	**44.7%**
Young Adult revenue	**$62,143**



5. Key Findings

Category Performance

Clothing generated **$104,264**, representing **44.7% of total revenue**. Clothing and Accessories together contributed **76.6% of revenue**, making them the strongest areas for inventory planning, merchandising, and cross-selling.

Category average purchase amounts remained close to the overall average. Clothing's revenue lead was therefore driven primarily by transaction volume rather than a substantially higher purchase value.

Customer Demographics

Young Adults ages 18–31 generated the highest age-group revenue at **$62,143**. Average purchase amounts were similar across age groups, indicating that differences in revenue were mainly associated with customer volume.

Discounts

Discounted purchases averaged **$59.28**, compared with **$60.13** for purchases without a discount. This snapshot does not show that discounts increased order value. Discount effectiveness should be evaluated using incremental conversion and profit rather than purchase amount alone.

Subscriptions and Repeat Purchasing

Subscribers represented **27% of customers**. They showed a slightly higher repeat-buyer share, but their current average purchase amount was nearly identical to that of non-subscribers. Subscription status alone was therefore not a strong signal of higher transaction value.

Reviews and Purchase History

Review ratings and previous-purchase counts had very weak relationships with current purchase amount. A higher rating or longer purchase history did not reliably correspond to a larger transaction in this dataset.

Payment and Shipping Preferences

Payment and shipping selections were broadly distributed. No single method dominated strongly enough to justify reducing customer choice. Maintaining flexibility may support convenience and customer satisfaction.

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6. Business Recommendations

1. Protect the Core Revenue Categories

Maintain strong availability and visibility for Clothing. Use Accessories as the primary cross-sell category through bundles, complementary product recommendations, and personalized campaigns.

2. Build Lifecycle-Based Engagement

Segment customers into New, Returning, and Loyal groups using previous purchases and stated purchase frequency.

- Use onboarding and second-purchase incentives for new customers.
- Send replenishment or reminder campaigns to returning customers.
- Offer recognition, early access, or exclusive benefits to loyal customers.

3. Replace Broad Discounts with Controlled Tests

Use customer-level holdout tests before expanding promotions. Measure incremental conversion, average order value, and contribution margin rather than assuming that discounted orders create additional value.

4. Strengthen the Subscription Proposition

Test clear benefits such as early product access, exclusive bundles, loyalty points, or shipping advantages. Track retention and incremental profit before increasing subscriber-acquisition spending.

5. Investigate Product Experience Gaps

Review high-volume products with below-average ratings—particularly Shirts and Blouses—to identify potential issues involving quality, fit, product descriptions, or fulfillment.

6. Preserve Checkout Flexibility

Continue offering multiple payment and shipping choices. Test personalized defaults to reduce checkout friction while preserving customer control.

7. Improve Future Data Collection

Add transaction date, sales channel, quantity, product cost, discount depth, returns, marketing exposure, and acquisition source. These fields will enable retention cohorts, customer lifetime value, profitability, channel performance, and stronger causal analysis.

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7. Limitations

The dataset is a cross-sectional snapshot and does not include transaction dates, product costs, quantities, returns, discount depth, marketing costs, or a true online-versus-store channel field. The results describe observed patterns and should not be interpreted as proof that any factor caused higher or lower spending.

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8. Next Steps

1. Build a dated transaction model for cohort and retention analysis.

2. Calculate customer lifetime value and contribution margin by segment.
3. Test promotional and subscription strategies using control groups.
4. Add channel and campaign attribution to future dashboards.
5. Monitor product ratings alongside returns and repeat-purchase behavior.

Conclusion

The strongest immediate opportunity is to protect the Clothing revenue base, use Accessories for cross-selling, and develop lifecycle-based customer engagement. Discounts and subscription initiatives should be optimized through controlled testing, while improved transaction and profitability data will allow the company to move from descriptive reporting to stronger predictive and causal decision-making.
